



Carnegie Mellon University
Master of
Software Engineering

History of Software Architecture

LG Architecture Training Program

Paulo Merson

Information sources

- M. Shaw and P. Clements. “The Golden Age of Software Architecture: A Comprehensive Survey” [Shaw06]
- P. Kruchten, H. Obbink & J. Stafford (2006). “The past, present and future of software architecture” [Kruchten06]
- G. Hohpe, I. Ozkaya, O. Zimmermann. “The Software Architect's Role in the Digital Age” [Hohpe16]
- G. Booch. “[History of Software Engineering](#)”. OOP 2021 keynote address.



Before *architecture* came *engineering*

“Software Engineering” was first
coined in 1968 at the NATO
conference...

... or so we believed.

The Study Group concentrated on possible actions which would merit an international, rather than a national effort. In particular it focused its attentions on the problems of software... The phrase ‘software engineering’ was deliberately chosen as being provocative, in implying the need for software manufacture to be based on the types of theoretical foundations and practical disciplines, that are traditional in the established branches of engineering.
[NATO68]





Margaret Hamilton

Worked at MIT and became the lead developer for the Apollo flight software, when she coined the term "Software Engineering" around 1963-64

Software *architecture* came later

- The first use of the term dates to 1969
- However, software architecture as a discipline started in 1991
 - “Software Architecture: Integrating Process and Technology” [Royce91]
- But important ideas paved the way...



Winston Royce

Before 1985: inspiring ideas about structures

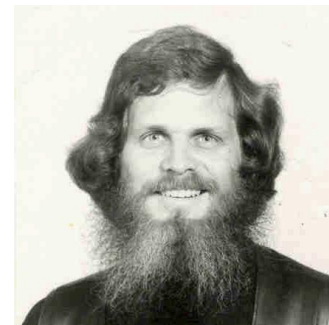
- *Producing the correct output is not enough; other qualities such as reliability and maintainability are also important, and can be achieved by careful structuring*
 - “The structure of the THE multiprogramming system” [Dijkstra68]
 - “On the criteria to be used in decomposing systems into modules” [Parnas72]
- *There are several structures with different purposes in software*
 - “On a ‘buzzword’: hierarchical structure.” [Parnas74]
- *Beyond procedure calls, how do we think about connecting modules*
 - “Programming-in-the-Large versus Programming-in-the-Small” [DeRemer76]



Edsger Dijkstra



Dave Parnas

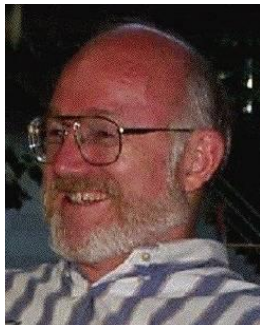


Frank DeRemer

1985-1994: basic research phase

- Design of specialized software structures for specific problems (e.g., avionics)
 - “The Modular Structure of Complex Systems” [Parnas85]
- Terms “view” and “style” borrowed from traditional building architecture
 - “Foundations for the study of software architecture” [Perry92]
- First catalogs of architecture styles (e.g., pipe and filter, blackboard)
 - “An Introduction to Software Architecture” [Garlan94]

Dewayne
Perry



Alexander
Wolf

Mary Shaw



David Garlan

1992-1996: concept formulation phase

- Architecture Description Languages (ADLs) (e.g., Meta-H, Rapide, Wright)
 - “A Classification and Comparison Framework for Software Architecture Description Languages” [Medvidovic97]
- Architecture views
 - “The 4+1 View Model of Software Architecture” [Kruchten95]
- Taxonomies of styles and first catalogs of architecture patterns
 - “Pattern-Oriented Software Architecture - A System of Patterns” [Buschmann96]
 - “Software Architecture: Perspectives on an Emerging Discipline” [Shaw96]
- Awareness of the relation between design decisions and quality attributes



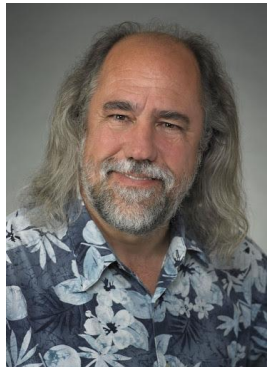
Philippe Kruchten



Frank Buschmann

1995-2000: development and extension phase

- IEEE TSE special issue on software architecture in 1995
- First WICSA conference in 1998 (now it's called ICSA)
- ACME is proposed as a unifying ADL
- Rational presents UML v0.9 to unify OOAD notations



Grady Booch



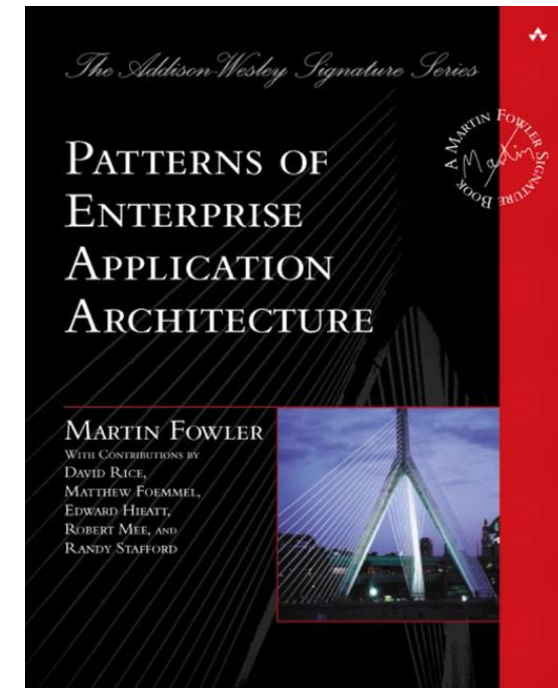
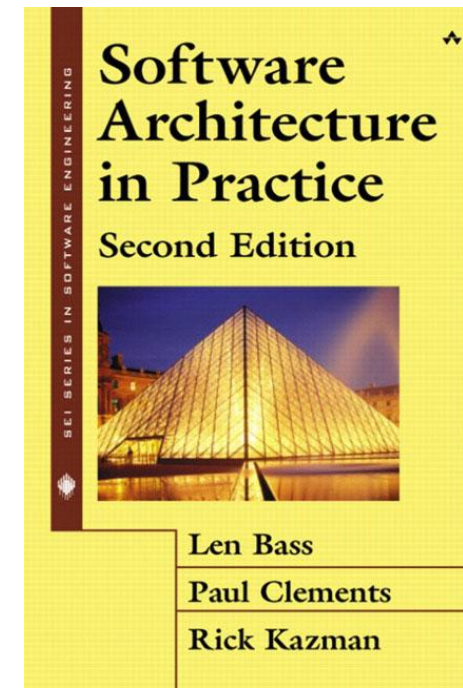
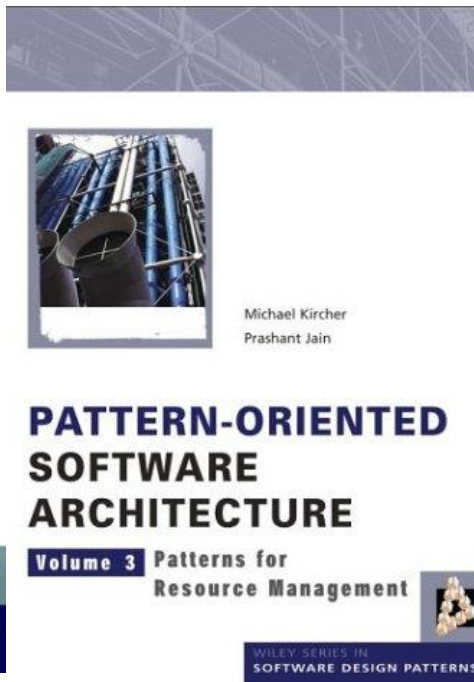
Ivar Jacobson



Jim Rumbaugh

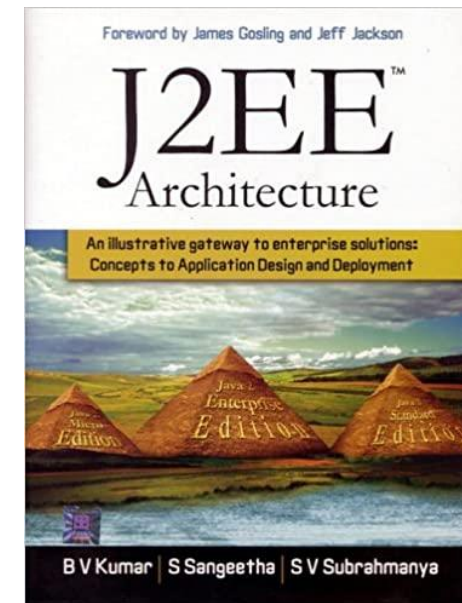
1996-2003: internal enhancement and exploration phase

- The Software Engineering Institute (SEI) creates ATAM
- Architecture patterns commonly used to guide software design
- Several books focused on practitioners



1998-2006: external enhancement and exploration phase

- UML becomes a *de jure* (OMG) and *de facto* standard
 - UML 2.0 published in 2004
- RUP and other architecture-driven development lifecycle models
- Architecture is the foundation for component-based software engineering
- “Pre-cooked, platform-specific architectures” (e.g., J2EE architecture)



2000-2012: popularization phase ⁽¹⁾

- Architecture patterns (e.g., client-server, multi-tier, SOA, MVC) become part of every developer's vocabulary
- IEEE 4271 and later ISO 42010 standards proposed [ISO 42010]
- Graduate and more recently undergrad courses focusing on architecture
- Professional courses and certifications on architecture
- Publications, conferences and professional entities



2000-2012: popularization phase ₍₂₎

- Software Architect #1 of Best Jobs in America (CNN Money, 2010)
- Bill Gates creates for himself the position of “Chief Software Architect”
- An assigned architect is commonly found in any mid-sized software project
- *The Architect* is portrayed as a God-like character by Hollywood



The KISS Architecture Model

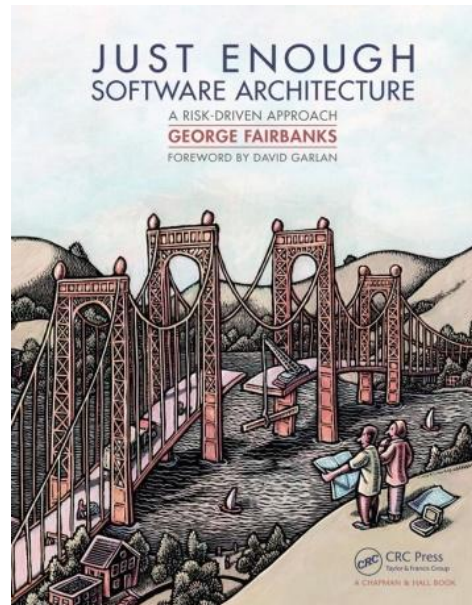
There are several architecture models with prescribed views and notations. The Keep It Short & Simple architecture model is different. We create pieces of documentation iff they benefit stakeholders. We do so using drawing tools, not modeling tools. We say no to BDUF and yes to Important Design Up Front. We follow 7 tips for creating diagrams that are expressive, not ambiguous, and help you to successfully understand and evolve them, and build a system from them. We complement the design diagrams that describe structure and behavior with ADRs.

2010-present: simplification phase

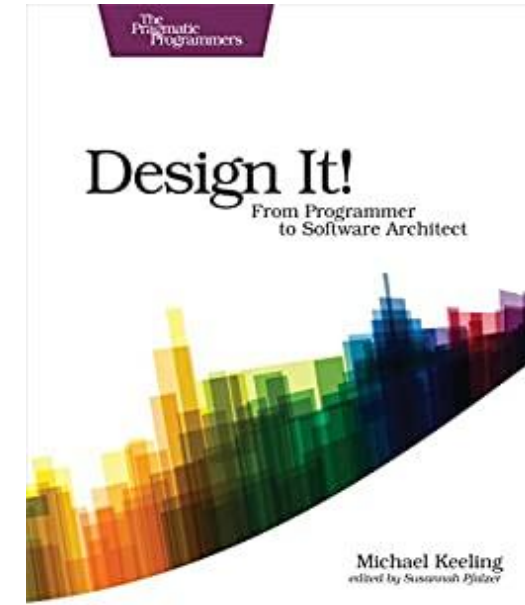
- Agile development and fast release cycles with no BDUF
 - Informal design notations win over ADLs
- Architecture lives in the code (or alongside the code)
 - Need for *architecturally-evident coding style*
 - Architecture Decision Records (ADRs)
- Frameworks and middleware platforms encapsulate key architecture decisions



Michael
Nygard



George
Fairbanks



Michael
Keeling



Questions?

Paulo Merson
pmerson@cmu.edu